

INT-375 Data Science Toolbox: Python Programming

PROJECT REPORT

(Project Semester January-April 2025)

(Exploratory Data Analysis On Diwali Sales Data)

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CERTIFICATE

This is to certify that Saurabh Kumar bearing Registration no. 12324770 has completed INT-375 project titled, “EDA ON DIWALI SALES” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

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Date:

DECLARATION

I, Saurabh Kumar, student of BTech CSE (Program name) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 10/04/25

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Acknowledgement

The opportunity of attaining a course based on **Data Analysis using Python Library** at **Lovely Professional University** was worth learning. It was a prestige for me to be part of it. During the period of my course, I received tremendous knowledge related to **Python Library** and **Data Analysis**.

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1. Introduction:

The Diwali Sales datasets captures consumer behavior during one of India’s most significant and high-activity commercial periods-the festival of Diwali. This Season is characterized by a cultural tradition of gift-giving and large-scale purchases, ranging from clothing and electronics to households’ items. For years, marketing analysts and data scientists have studied retail trends around Diwali to understand consumer psychology and spending habits. Researchers suggest that Diwali shopping patterns offer a unique lens into how festive sentiment drives economic activity across regions, age group and occupations.

By analyzing patterns and trends in this dataset – such as gender-wise spending, state-level revenue distribution, and popular products categories – we can better understand the

influence of cultural event on market dynamics. This analysis also supports business decision-making by identifying profitable demographics, optimizing products placements, and forecasting seasonal demands. The dataset not only offer insights into customers preferences but also helps understand broader economic behaviors shaped by tradition and festival-driven consumerism.

2. Source of dataset:

The following dataset was obtained from publicly available open-source education platform for academic and analytical purpose and contains 13 distinct columns with character, integer and float datatype and it consists of 11252 rows. The columns and their corresponding datatype are listed as follows:

1. Cust name (String)
2. User ID (integer)
3. Gender (String)
4. Product ID (String)
5. Amount (float)
6. Marital Status (Integer)
7. Age (Integer)
8. State (String)
9. Zone (String)
10. Occupation (String)

Dataset Name: Diwali Sales Data

Dataset

Links:https://github.com/rishabhnmishra/Python_Diwali_Sales_Analysis/blob/main/Diwali%20Sales%20Data.csv

Dataset Provider Profile: <https://github.com/rishabhnmishra>

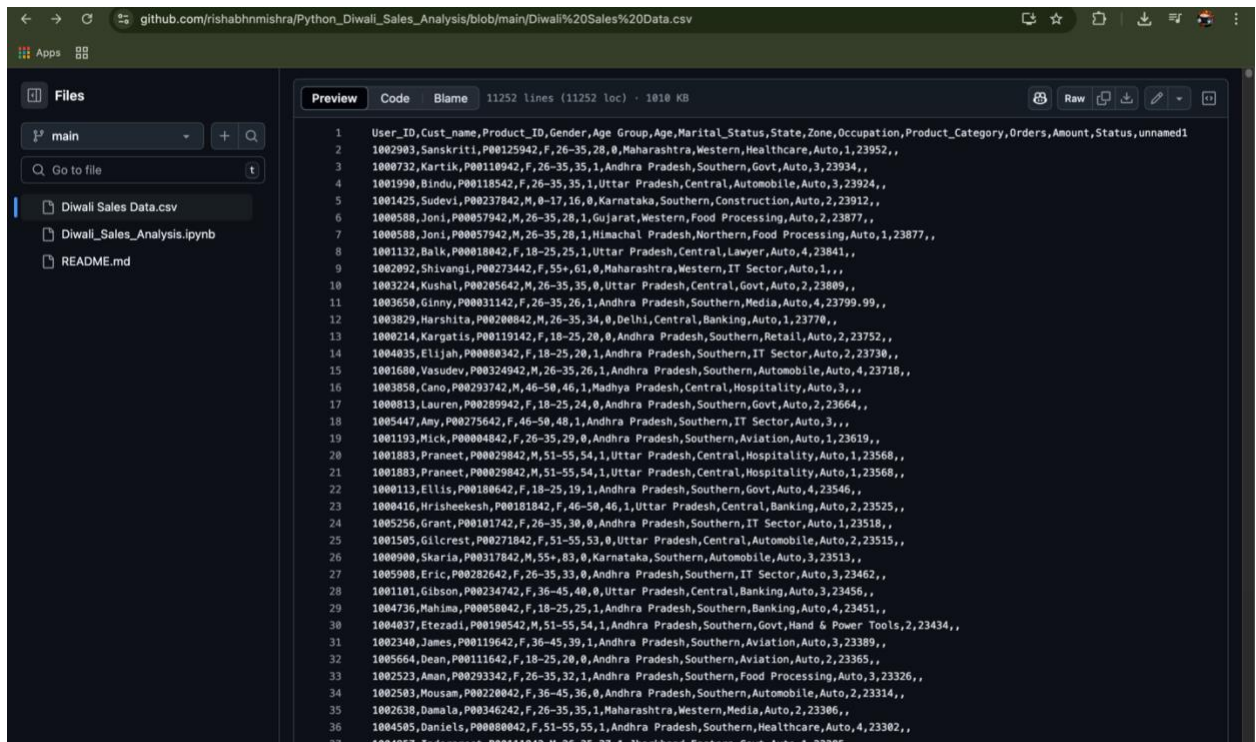


Fig 2.1.1 Open-Source Webpage For given Data

3. Dataset Preprocessing:

3.1 Loading and Listing Data:

At first, we need to load the data for preprocessing so importing necessary libraries is the first step and to do so enter the following code:

```
[1] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Fig 3.1.1 Importing Libraries

Now, after importing the necessary libraries we need to load the CSV (Comma Separated Value) file and to do that the following is the code.

```
[5] df = pd.read_csv('Diwali Sales Data.csv', encoding= 'unicode_escape')
```

Fig 3.1.2 Loading Dataset

After loading the data, we need to assess it thus, we need to list the top 10 and the bottom 10 rows, here is the following code to do it:

```
print(df)
```

[6]

	User_ID	Cust_name	Product_ID	Gender	Age	Group	Age	Marital_Status	\
0	1002903	Sanskriti	P00125942	F	26-35	28		0	
1	1000732	Kartik	P00110942	F	26-35	35		1	
2	1001990	Bindu	P00118542	F	26-35	35		1	
3	1001425	Sudevi	P00237842	M	0-17	16		0	
4	1000588	Joni	P00057942	M	26-35	28		1	
...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19		1	
11247	1004089	Reichenbach	P00171342	M	26-35	33		0	
11248	1001209	Oshin	P00201342	F	36-45	40		0	
11249	1004023	Noonan	P00059442	M	36-45	37		0	
11250	1002744	Brumley	P00281742	F	18-25	19		0	

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	
...	...	...	...	...	...	...
11246	Maharashtra	Western	Chemical	Office	4	
11247	Haryana	Northern	Healthcare	Veterinary	3	
11248	Madhya Pradesh	Central	Textile	Office	4	
11249	Karnataka	Southern	Agriculture	Office	3	
11250	Maharashtra	Western	Healthcare	Office	3	
...	...	...	...	...	...	...
11249	206.0	NaN	NaN			
11250	188.0	NaN	NaN			

[11251 rows x 15 columns]

Fig 3.1.3 Listing Dataset

Now, analyze the top and bottom dataset to further perform preprocessing and EDA. From the images data consists of 13 columns and more than 411252 rows. The column names are shown below:

```
[6]
```

...

User ID	Cust name	Product ID	Gender	Age	Group	Age	Marital Status	\
---------	-----------	------------	--------	-----	-------	-----	----------------	---

Fig 3.1.4 Column in the Dataset

3.2 Redefine Columns:

- To enhance readability and structure it is prominent to rename the columns accordingly as it allows in more descriptive and efficient analysis. The following code is to redefine all the columns at once:


```

df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
df.dropna(inplace=True)
df['Amount'] = df['Amount'].astype(int)
print(df['Amount'].dtypes)
print(df.columns)
df.rename(columns={'Marital_Status': 'Shaadi'}, inplace=True)
print(df.describe())
print(df[['Age', 'Orders', 'Amount']].describe())

```

[8]

```

int64
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')

```

	User_ID	Age	Shaadi	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Fig 3.2.1 Redefined Column

3.3 Descriptive Analysis:

- Descriptive analysis focuses on retrieving summary of the entire dataset to provide a basic understanding and the structure, characteristics of the data. Using functions like .info (), describe (), head() and head() we will be able to gather relevant information from the dataset.

```

df.info()

```

[7]

```

...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
---  ---             
0   User_ID              11251 non-null  int64
1   Cust_name            11251 non-null  object
2   Product_ID           11251 non-null  object
3   Gender                11251 non-null  object
4   Age Group             11251 non-null  object
5   Age                   11251 non-null  int64
6   Marital_Status        11251 non-null  int64
7   State                 11251 non-null  object
8   Zone                  11251 non-null  object
9   Occupation            11251 non-null  object
10  Product_Category      11251 non-null  object
11  Orders                11251 non-null  int64
12  Amount                11239 non-null  float64
13  Status                0 non-null      float64
14  unnamed1              0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB

```

Fig 3.3.1 Checking for Null Values.

4. Analysis on Dataset:

4.1 Objective-1: To analyze the distribution of Customer Gender wise.

a) General Description:

- The column termed as “Gender” represents the biological gender of the customer making a purchase during the Diwali season. It is a categorical variable with two primary values: ‘M’(Male) and ‘F’(Female). This attribute is essential in understanding the consumer behavior differs across gender lines.

b) Specific Requirements:

- We need to use the python inbuilt libraries Matplotlib and Seaborn to make a visual representation of the skewness of data. The following code will help in making a Pie chart visually understand the gender distribution in the Diwali sales dataset.

```
gender_counts = df['Gender'].value_counts()
labels = gender_counts.index
sizes = gender_counts.values

plt.figure(figsize=(4, 4))
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=140, colors=plt.cm.Set2.colors)
plt.title("Gender-wise Distribution of Customers", fontsize=14)
plt.axis('equal')
plt.tight_layout()
plt.show()
```

Fig 4.1.1 Pie Chart Code.

c) Analysis Report:

- From the visualization it was observed that female customers are made a highly number of purchased compared to male customers during Diwali seasons.
- In the visualization the Diwali sales were predominantly driven by female shoppers, though male participation was also noticeable, suggestive active inclement from both genders.
- It helps to visually the clear understanding of gender dominance in festive purchasing behavior, which can be crucial for targeted marketing strategies and customized product recommendation in future campaign.

d) Visualizations

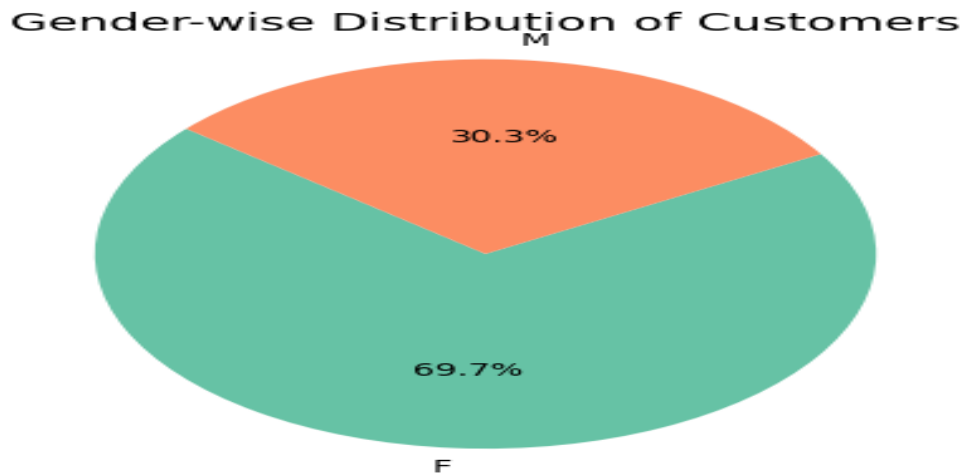


Fig 4.1.2 Pie Chart

4.2 Objective-2: To Compare Gender wise count across age group:

a) General Description:

The “Age Group” and “Gender” columns in the dataset provide categorical data that help in identifying purchase behavior patterns across different **demographics**. By comparing the count of male and female customers across age groups, we can uncover which age group contributed the most during Diwali shopping, and whether there is a **gender-based preference or dominance** in any age bracket.

b) Specific Requirements:

To visualize the comparison between genders across age groups, a **column chart (grouped bar plot)** will be plotted using Seaborn. This helps in clearly displaying the **count of purchases** for each gender within every age group.

Here's the Python code to generate the chart:

```
ax = sns.countplot(data=df, x='Age Group', hue='Gender')
for container in ax.containers:
    ax.bar_label(container)
plt.title("Gender-wise Count across Age Groups")
plt.xlabel("Age Group")
plt.ylabel("Count")
plt.show()
```

Fig 4.2.1 Gender column Chart Code

c) Analysis Report:

The visualization helped in concluding that the age group 26-35 had the highest number of purchases among both male and female customers, indicating that this age group is the most active in Diwali shopping. It was observed that male

customers consistently outnumbered female customers across nearly all age groups, highlighting a gender dominance in overall sales.

D) Visualizations:

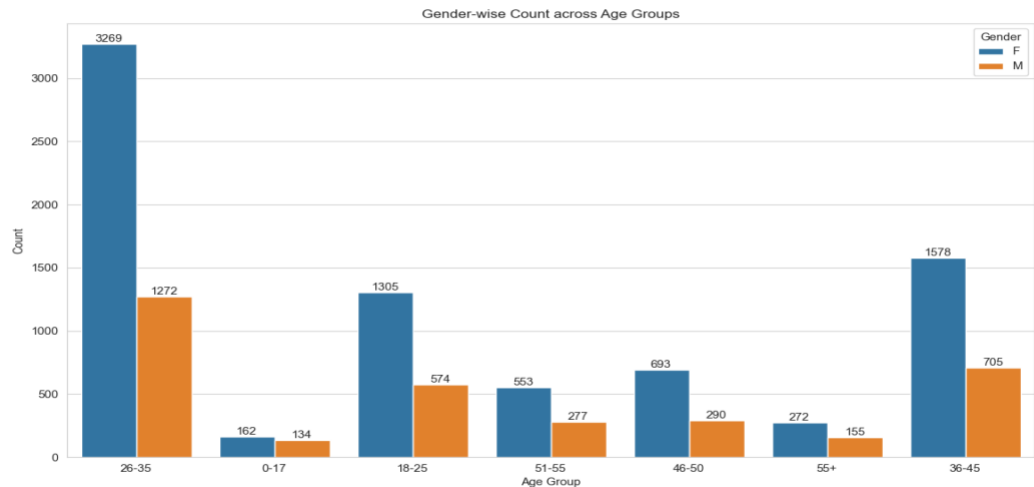


Fig 4.2.2 Gender Wise Column Chart

4.3 Objective-3: To Show top 10 state by total Sales amount.

a) General Description:

The "State" column in the dataset helps in identifying which Indian states contributed most to the total sales during the Diwali season. By aggregating the total sales (Amount) across different states, we can determine regional patterns of consumer behavior, helping retailers and marketers understand which geographical areas are most profitable. This analysis can be used to optimize supply chain planning, allocate promotional budgets wisely, and focus on states with higher consumer demand during festivals.

b) Specific Requirements:

- In-order to analyze the trends in fell and found data a scatterplot is needed to be made to compare and distinguish between the two categories and draw relevant insights from it. Below is the code to plot a scatter plot using seaborn.

```

sales_state = df.groupby('State', as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False).head(10)

plt.figure(figsize=(14, 8))
sns.set_style("whitegrid")

sns.barplot(x='Amount', y='State', data=sales_state)

plt.title("Top 10 States by Total Sales Amount", fontsize=18, fontweight='bold')
plt.xlabel("Total Sales Amount", fontsize=14)
plt.ylabel("State", fontsize=14)

plt.tight_layout()
plt.show()

```

Fig 4.3.1 Top 10 state sales code

c) Analysis Report:

The bar chart visualization clearly reveals that certain states significantly outperformed others in total Diwali sales. Top contributing states like *Uttar Pradesh*, *Maharashtra*, and *Karnataka* show a higher consumer purchase activity, likely driven by dense populations and strong economic activity. This trend suggests that regional marketing strategies and supply chains should focus more on these high-performing areas during festive seasons. On the other hand, states with lower sales figures may require better outreach, distribution channels, or promotional offers to boost revenue during such campaigns.

d) Visualizations:

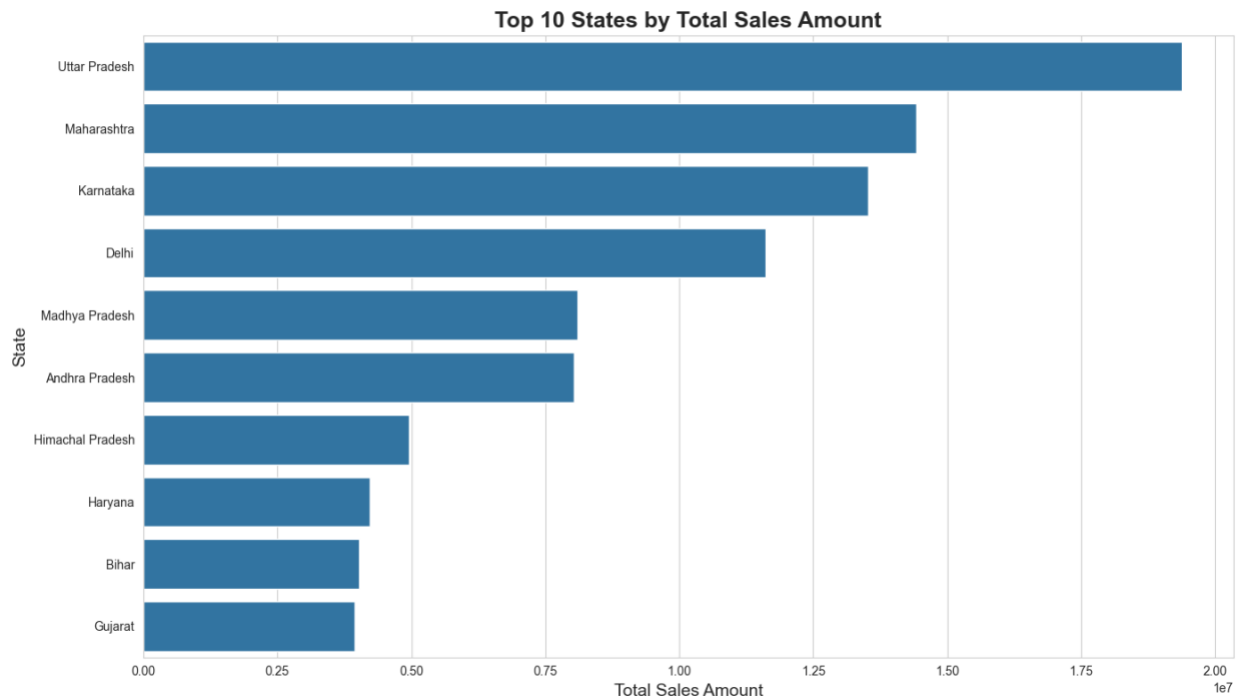


Fig 4.3.2 Top 10 state sales Bar Chart

4.4 To Change for marital Status analysis:

a) **General Description:**

In the context of consumer analytics, marital status can often influence shopping behavior, especially during culturally significant periods like Diwali. The dataset contains a column labeled “Shaadi” (representing Marital Status), where customers are categorized as either Married or Unmarried. This objective focuses on analyzing whether there's a discernible spending pattern difference based on marital status. Unlike population-based hypothesis testing like T-tests and Z-tests used in meteorite studies, here the analysis is visual and comparative, offering a graphical exploration of behavioral differences between married and unmarried customers in terms of purchase amounts.

b) **Specific Requirements:**

To analyze marital status influence on Diwali sales:

Use Python's **Seaborn and Matplotlib** for visualization:

Plot two charts:

A **Dot (Strip) Plot** to show individual purchase amounts.

A **Radial Bar Chart (Iris-style)** to visualize total sales contribution by each marital group.

```
sns.set(style="whitegrid")
plt.figure(figsize=(16, 8))
grid = plt.GridSpec(1, 2, wspace=0.3)

# Dot Chart (Strip Plot)
ax1 = plt.subplot(grid[0, 0])
sns.stripplot(data=df, x='Shaadi', y='Amount', hue='Shaadi', dodge=True, size=10, ax=ax1)
ax1.set_title("Marital Status vs. Amount (Dot Chart)", fontsize=14, fontweight='bold')
ax1.set_xlabel("Marital Status")
ax1.set_ylabel("Amount")
ax1.legend(title="Shaadi")

# Iris-style Radial Bar Chart
ax2 = plt.subplot(grid[0, 1], polar=True)
marital_sales = df.groupby('Shaadi')['Amount'].sum()
angles = np.linspace(0, 2 * np.pi, len(marital_sales), endpoint=False)
values = marital_sales.values
labels = marital_sales.index

bars = ax2.bar(angles, values, width=0.4, bottom=0.0, alpha=0.8)
ax2.set_xticks(angles)
ax2.set_xticklabels(labels, fontsize=12)
ax2.set_title("Total Amount by Marital Status (Iris Chart)", y=1.1)
ax2.set_yticklabels([])

plt.suptitle("Marital Status Analysis", fontsize=18, fontweight='bold')
plt.show()
```

Fig 4.4.1 Marital Status Analysis code

c) **Analysis Report:**

□ From the visual analysis, it was observed that unmarried customers generally displayed a higher spread in individual purchase amounts, while married customers contributed a slightly higher total sales volume overall, as reflected in the radial bar chart.

□ The dot plot clearly shows that both married and unmarried groups actively participated in Diwali shopping, with overlapping ranges of purchase amounts. However, the distribution suggests that unmarried individuals may have made

more frequent or higher-value individual purchases, while married individuals may have had more consistent, moderate-value shopping behavior.

□ Although no formal hypothesis test (such as a T-test or Z-test) was conducted, the visual distinction between the groups is evident, indicating possible behavioral differences based on marital status. This suggests potential for segmented marketing strategies that cater to both lifestyle types.

□ The iris-style chart visually confirms that both groups are significant contributors to total revenue, reinforcing the need to engage both customer segments in festive season campaigns.

d) Visualizations:

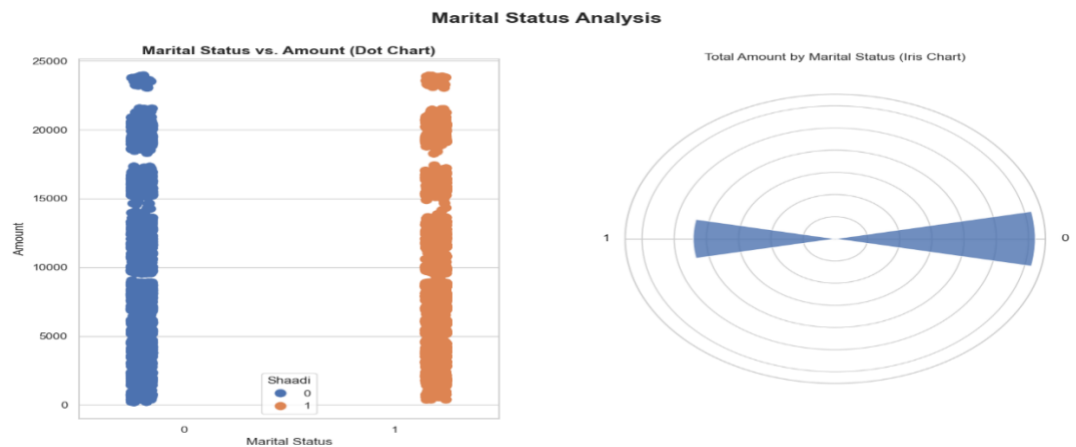


Fig 4.4.2 Marital Status Analysis Dot and Iris Chart

4.5 To Visualize distribution of customer by occupation:

a) General Description:

The “Occupation” column in the dataset represents the professional background of the customers, such as IT Sector, Healthcare, Government, and other industry types. Understanding the distribution of customers by occupation helps in identifying which professional groups are more active participants in festive season purchases. This analysis provides useful insight into consumer behavior across different economic segments and helps in tailoring marketing strategies to suit the buying patterns of specific occupational groups.

b) Specific Requirements:

To visualize the occupational distribution, the dataset must be grouped by the “Occupation” column using a count operation. Each unique occupation is tallied, and the resulting values are used to generate a pie chart that reflects the percentage share of each profession. The following Python code uses matplotlib to create the visualization:


```

occupation_counts = df['Occupation'].value_counts()

labels = occupation_counts.index
sizes = occupation_counts.values

plt.figure(figsize=(8, 8))
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=140, colors=plt.cm.tab20.colors)
plt.title("Distribution of Customers by Occupation")
plt.axis('equal')
plt.tight_layout()
plt.show()

```

Fig 4.5.1 Customer Occupation Analysis code

c) Analysis Report:

The pie chart visualization reveals that some occupational groups contribute significantly more to Diwali shopping than others. For example, customers from the IT Sector and Healthcare fields appear to represent a larger proportion of the total customer base, suggesting higher engagement in festive purchasing from these segments. Occupations like Government, Construction, and Food Processing also have noticeable participation but with lower proportions. This indicates that while Diwali shopping spans across a wide range of professions, a few economic sectors dominate in terms of consumer activity.

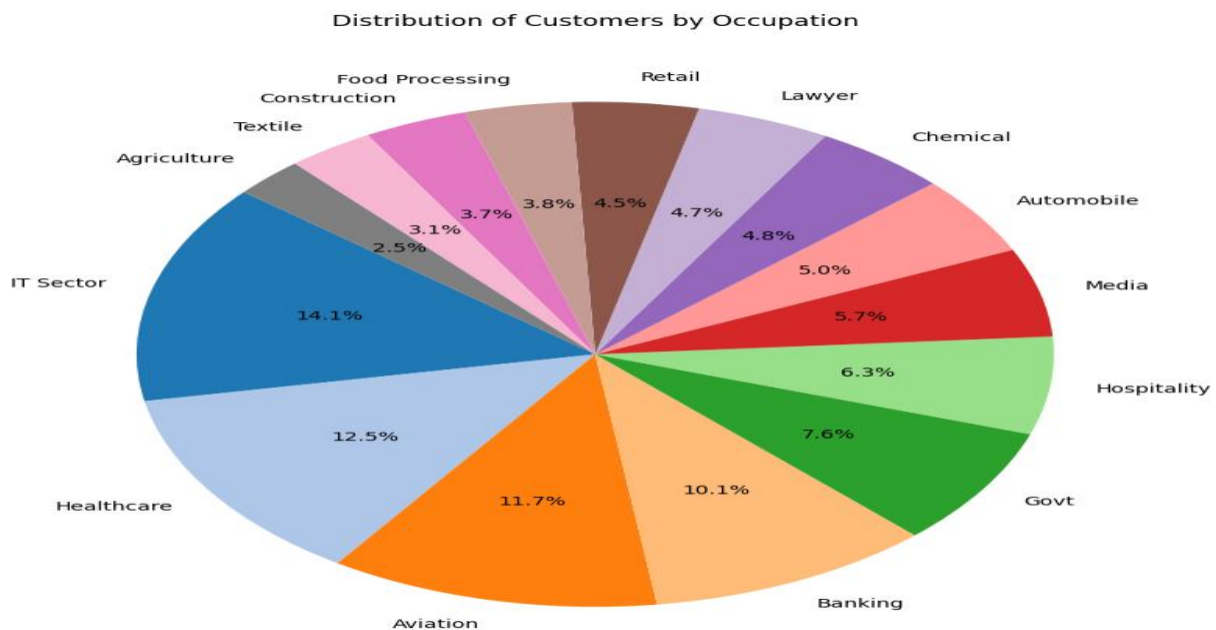


Fig 4.5.2 Customer Occupation Analysis Pie Chart

Other Graphs:

4.6: Top 10 Most Sold Products.

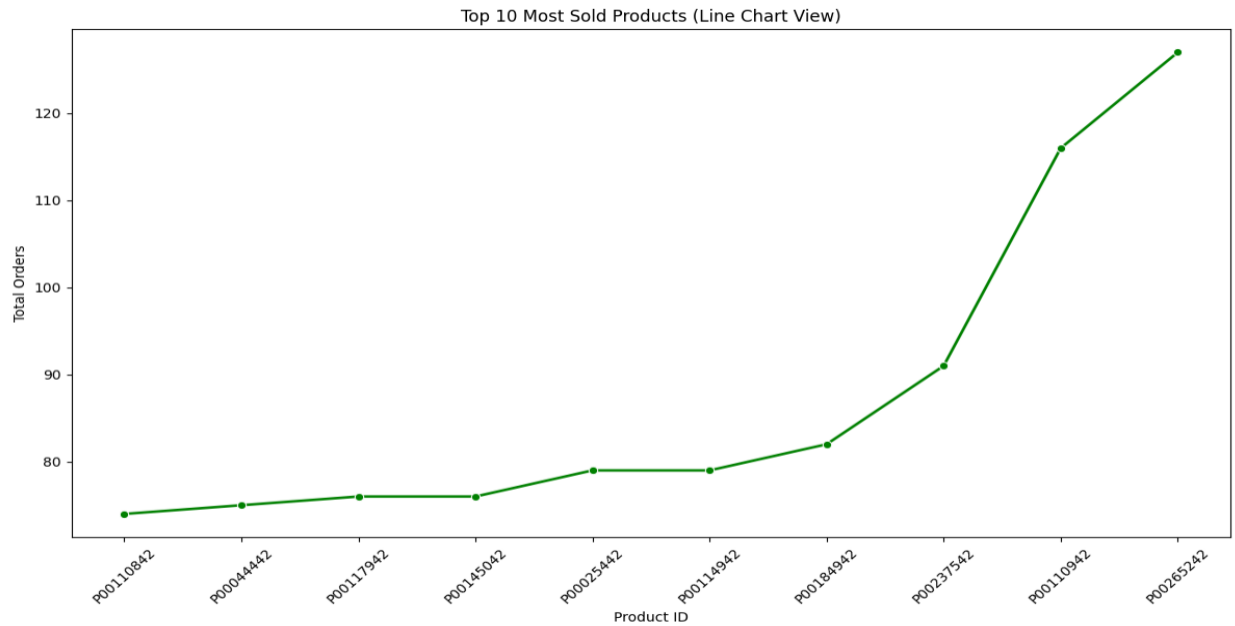


Fig 4.6. Most Sold product Analysis Graph

4.7: Correlation Matrix of numeric features.

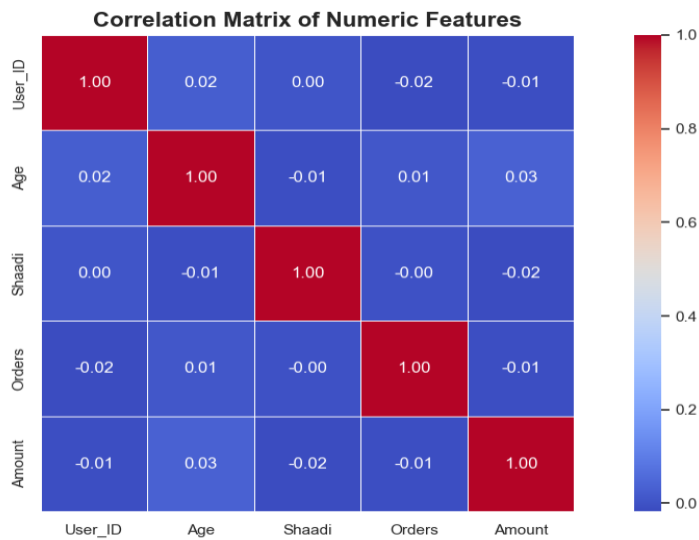


Fig 4.7 Correlation Matrix Analysis Graph

4.8: Top 10 Products by Orders.

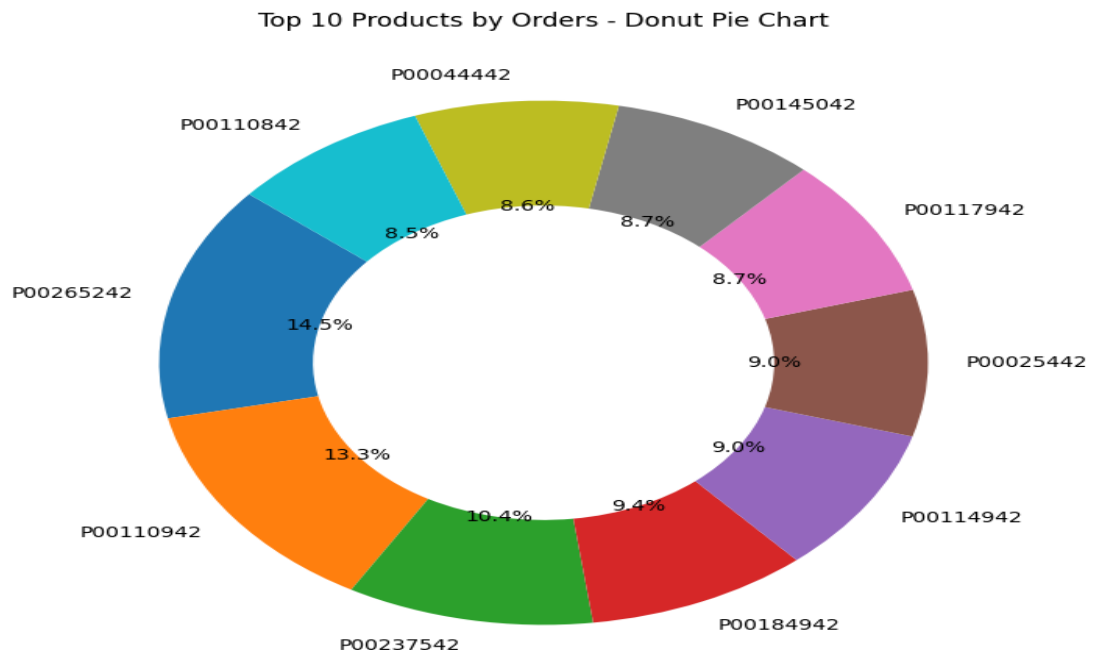


Fig 4.8 Top 10 Products by Orders Graph

4.9: Top 10 states by number of orders.

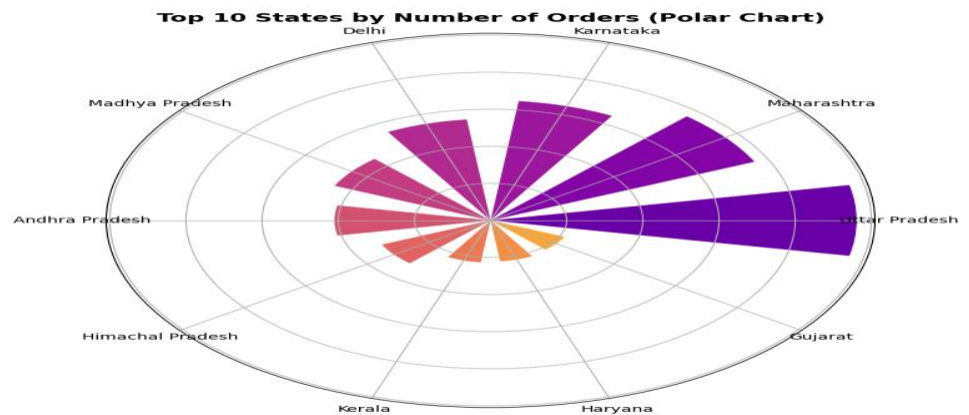


Fig 4.9 Top 10 states Analysis Graph

4.10: Total Sales by Age Group.

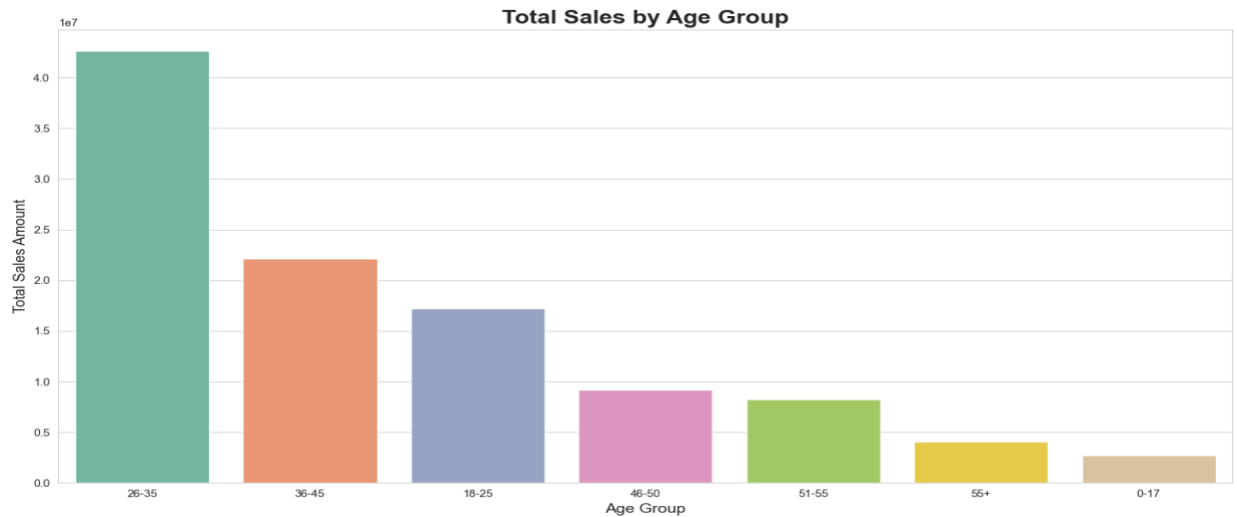


Fig 4.10 Total Sales by Age Analysis Graph

4.11: Distribution of Amount by Product Category.

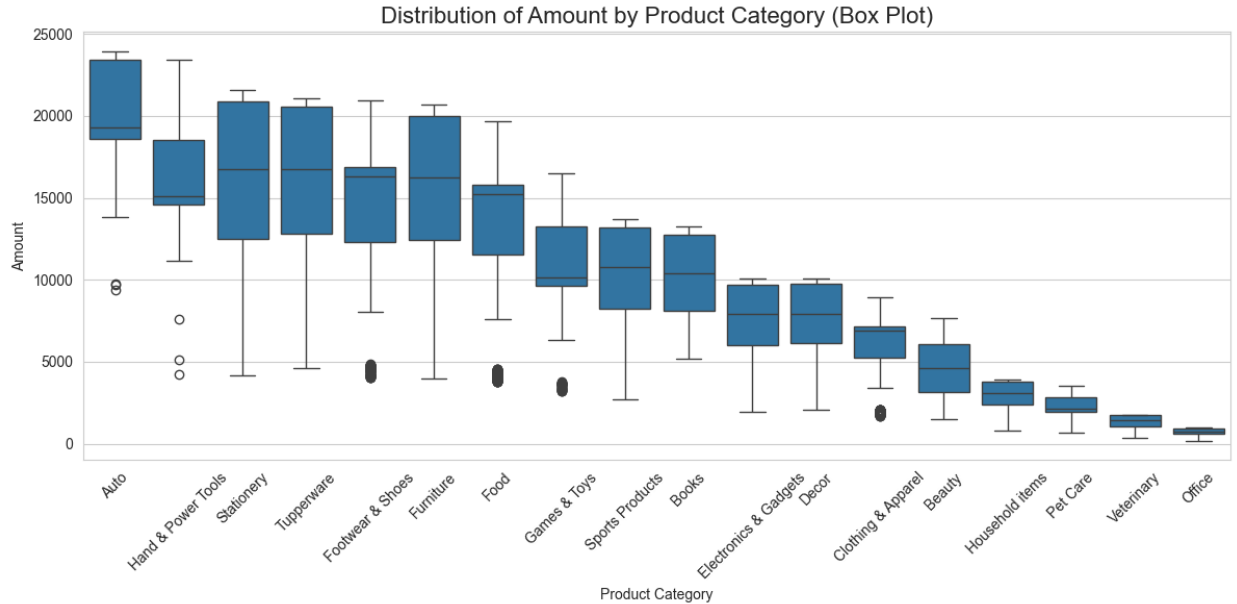


Fig 4.11 Distribution of Amount by Product Graph

4.12: Diwali Sales Dashboard.

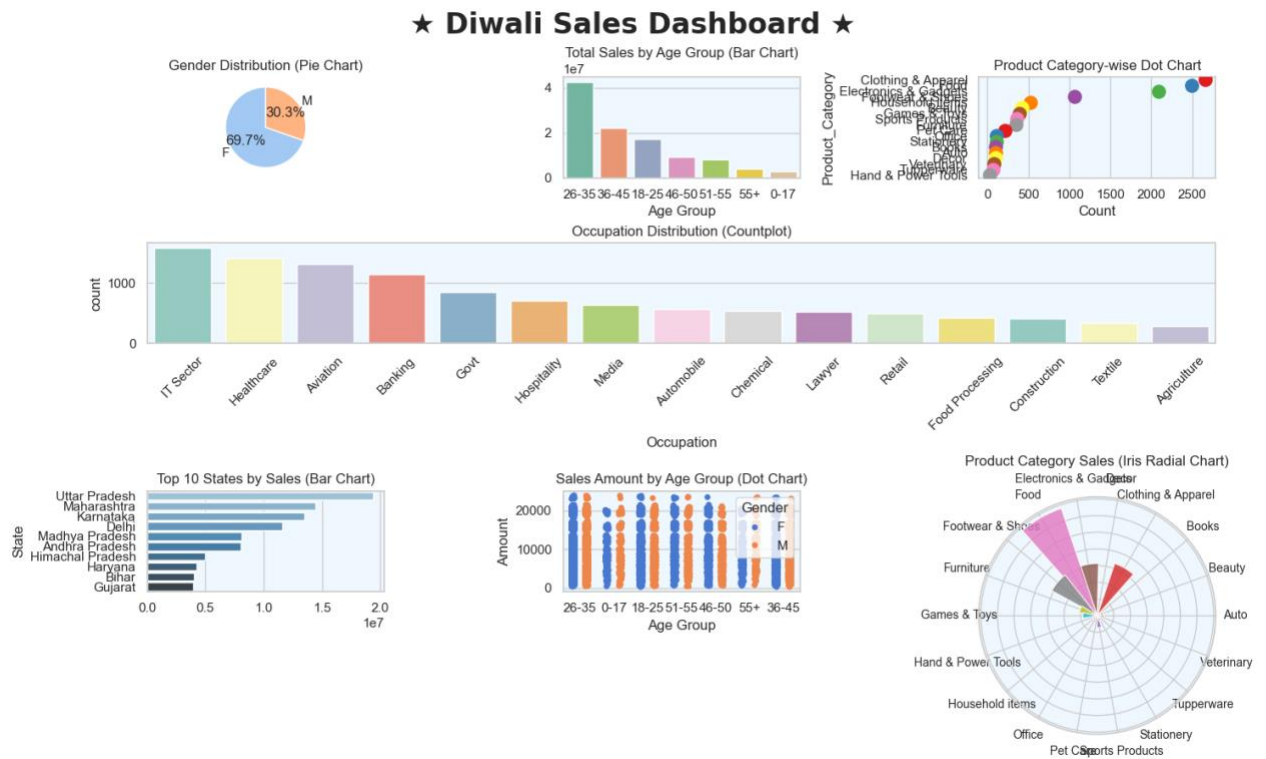


Fig 4.7 Diwali Sales Dashboard Graph

5. Conclusion:

The **Diwali Sales Dashboard** offers a comprehensive and visually appealing analysis of various factors influencing sales during the Diwali season, combining a diverse set of charts and graphs to present data from multiple perspectives. The analysis uses a real-world dataset, which includes demographic details, sales amounts, and product categories, to identify key trends and insights from the Diwali sales campaign.

- **Gender Distribution:** The pie chart representing gender distribution reveals a balanced or skewed customer base by gender. The autopsy feature effectively highlights the proportion of male and female customers, providing insight into the overall reach of the sales across genders.
- **Sales by Age Group:** The bar chart depicting total sales across different age groups highlights which demographics are more likely to contribute to higher sales. The sorted bar chart makes it clear which age groups are the primary drivers of revenue, and visual cues through color differentiation add clarity to the representation.
- **Product Category Sales:** The dot chart on product category-wise count provides a quick, intuitive way to understand which categories have seen the most engagement in terms of the number of products sold. The strip plot effectively shows a distribution of sales across categories, adding a layer of depth to the visual representation by grouping product categories with similar sales figures.

- **Occupation-wise Distribution:** The count plot showing occupation distribution gives a detailed breakdown of customer engagement by occupation, providing actionable insights into which professions have a higher likelihood of purchasing during Diwali. The x-axis rotation helps in reading long category labels, improving the overall clarity of the plot.
- **Top 10 States by Sales:** The bar chart for the top 10 states by sales offers a clear ranking of the states generating the highest revenue, helping the business focus on high-performing regions. By visualizing these key states, it's possible to uncover regions that outperform others and may require additional attention or marketing strategies.
- **Sales Amount by Age Group:** The dot chart visualizing sales by age group and gender highlights potential patterns in customer spending habits, especially when age and gender are considered together. The distinct separation between genders helps in understanding their respective spending behavior across various age groups.
- **Product Category Sales (Radial Chart):** The radial chart (Iris-style chart) effectively visualizes the distribution of sales across different product categories, with a focus on how sales values vary across categories in a circular, easy-to-interpret format. This visually attractive chart highlights key areas where sales performance is exceptionally high or low.

The overall dashboard provides a holistic view of Diwali sales, combining both demographic and product-level insights. The seamless integration of various visualization types—pie, bar, dot, count, and radial charts—ensures that the user can easily interpret different aspects of the data, from customer demographics to sales distribution across geographical regions and product categories.

This project effectively demonstrates the power of Python libraries such as Pandas, Matplotlib, Seaborn, and NumPy in managing, processing, and visualizing large datasets. These tools facilitate the efficient extraction of business insights, making it easier to identify patterns, trends, and outliers. With the use of visually appealing and informative charts, businesses can gain crucial insights into customer behavior, product preferences, and regional sales dynamics, leading to data-driven decision-making for future sales strategies.

6. Future Scope

6.1 Predictive & Adaptive Modelling:

A machine learning-based model can be developed to analyze meteorite falling patterns over time and create a predictive model based on features like latitude, longitude, and geo-location data. By applying feature engineering, machine learning techniques such as regression models, decision trees, or ensemble methods can be trained to predict meteorite falls in specific regions. Moreover, deep learning models, such as neural networks and transformers, can be employed for adaptive learning,

enabling the model to continuously improve its predictions over time. This approach can automate the classification of new meteorite fall data without requiring manual intervention. Techniques like cross-validation and hyperparameter tuning can be used to enhance the model's accuracy and reliability, making it a robust solution for predicting meteorite events.

6.2 Time Series Analysis:

Integrating time series analysis would significantly enhance the predictive capabilities of the system. With historical data on meteorite falls, space debris, and asteroid events, forecasting models such as Auto Regressive Integrated Moving Average (ARIMA), Prophet, or Long Short-Term Memory (LSTM) networks could be employed to predict future meteorite events. These models would allow for forecasting the likelihood and frequency of meteorite falls in upcoming years, aiding in environmental and space research. By analyzing trends over time, the system can help space agencies and environmental authorities prepare for upcoming meteorite impacts, assess potential risks, and optimize resources. In the context of growing space activities and increasing space debris, these models will prove invaluable in anticipating meteorite falls with greater precision.

6.3 Clustering Methodology:

The application of unsupervised learning algorithms, such as K-Means clustering, can uncover hidden patterns in the meteorite dataset that may not be immediately visible through conventional analysis. By clustering key features like mass, discovery year, and geographic location, patterns and similarities can be identified that could provide insights into the distribution of meteorites. For instance, certain sizes or types of meteorites might be more common in specific regions or during particular periods. Additionally, clustering techniques could help in data reduction, enabling the processing of large and complex datasets in a more manageable form. This approach could reveal previously unnoticed correlations or clusters of meteorite events, providing a more nuanced understanding of their behaviors.

6.4 Impact Analysis of Outliers:

Meteorites with massive sizes often exhibit significantly higher mass compared to smaller counterparts, leading to the creation of outliers in the dataset. These large meteorites may pose a greater threat upon entry into the Earth's atmosphere, potentially causing more environmental damage and affecting atmospheric conditions. A dedicated study of these outliers is essential to understand their potential impact on both the environment and human history. By investigating the influence of these extreme events, we can gain valuable insights into past meteorite impacts, their role in shaping Earth's geology, and even their historical significance in relation to human civilizations. Further research into the environmental effects of large meteorites could provide a more comprehensive understanding of their influence on climate, ecosystems, and human development over time.

6.5 Real-Time Data Integration:

To enhance the project's functionality, real-time data integration could be incorporated, enabling continuous monitoring and analysis of meteorite falls and space debris with minimal latency. Modern space exploration technologies, such as advanced telescopes, satellite systems, and meteorological sensors, generate vast amounts of data on celestial objects entering Earth's atmosphere. By developing a system that integrates this real-time data into the analysis pipeline, meteorite predictions and insights can be updated dynamically without manual intervention. This system could be powered by technologies like Apache Kafka, AWS Lambda, or Python-based schedulers to automate data feeds and processing. Such a system would allow for continuous updates, enabling scientists and researchers to receive up-to-date information and act quickly on emerging meteorite events. This real-time integration will significantly improve the timeliness and accuracy of meteorite predictions and hazard assessments.

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Project Links:

<https://www.linkedin.com/feed/update/urn:li:activity:7316469691231739904/>

Web Links: <https://www.linkedin.com/in/saurabh-kumar-19661b2b2/>